History-Preserving Quadratic Normal Forms for Circular Coalescence

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Abstract

A terminal placed partition does not determine the mergers that produced it. For a circular packet system generated by positive rotation and the adjacent collapse $d(1) = 0$, we characterize the complete effective-fusion logs and terminal placements jointly attainable from any initial placement of t disjoint named packets on n sites. Every ordered cyclic-neighbor merger schedule and compatible literal endpoint have a common continuous execution. We give a representative of length at most $\frac{7}{2}n(n-1)$ preserving the complete log and endpoint, with explicit routing budgets. At terminal rank s, the joint objects number $(t!/s!)s\binom{n}{s}$. For a specified singleton-start all-right history, the exact minimum is $\frac{3}{2}n(n-1)-1$; its endpoint alone has a linear-length realization. Thus retaining the prescribed merger history can raise the cost of that endpoint from linear to quadratic.

Keywords: circular coalescence, fusion history, transformation semigroups, constructive normal forms, word length

1 Introduction

An execution of a merging system has several possible outputs. Its final transformation records where the atoms went. Its terminal placed partition records which atoms now travel together and where their packets are. A complete effective-fusion log additionally records the actual ordered parents of every merger and the order in which the mergers occurred. Executions with the same endpoint can have different logs, and prescribing a log can change the shortest realization substantially.

We consider the literal alphabet consisting of a positive rotation p on an n-cycle and one adjacent idempotent d, which sends site 1 to site 0 and fixes all other sites. Every occurrence of either letter has unit cost. The initial packets may be arbitrary nonempty, disjoint atom sets at arbitrary distinct sites. Their initial physical cyclic order, rather than a numerical order on the atoms, supplies the relevant geometry.

The construction realizes the prescribed merger schedule on the current placement, then uses fusion-free routing to reach its compatible endpoint. Section 4 gives the joint realization and quadratic bound; Section 5 counts the logs and their placement fibres. Sections 6–7 compare the exact cost of one prescribed history with the cost of its endpoint alone.

These questions are adjacent to the study of transformation semigroups generated by a group and an idempotent (McAlister, 1998), and generating ranks for endomorphisms of finite circular orders (Levi and Mitchell, 2006). Cameron, Castillo-Ramirez, Gadouleau and Mitchell study $\ell(D, \alpha)$, the minimum number of digraph-arc generators expressing a transformation α (Cameron et al., 2017, Section 1). Each available arc $a \to b$ is a defect-one idempotent letter of unit cost. Here only the fixed letters p, d have unit cost. For example, the circular adjacent arc $b+1 \to b$ is realized chronologically by $p^{[-b]_n} d p^{[b]_n}$, whose expanded length is $1 + [-b]_n + [b]_n$: it is $n+1$ when $b \neq 0$, and one when $b = 0$. The displayed conjugate is not an additional unit-cost letter.

Their target is an endpoint transformation. From our singleton start, the literal endpoint determines the induced transformation on atoms, but equality of the complete log and endpoint is strictly finer than endpoint equality alone, as Section 7 demonstrates. More generally, our target from any fixed packet start P_0 is the pair consisting of its complete ordered fusion log and its terminal placed partition.

Earlier registered-summary classifications retain an origin event, later mergers involving a selected carrier, and its terminal placed partition (Chen, 2026a,b). Here the recorded object is the complete effective-fusion log and endpoint, with no carrier selected in advance. Projecting that log onto a selected carrier yields short representatives for attained registered summaries.

2 Placements and Complete Logs

Fix $n \geq 3$ and $Q = \mathbb{Z}/n\mathbb{Z}$. Let Ω be a finite nonempty atom set, independent of Q. A placed partition is a map $P : Q \to 2^{\Omega}$ whose nonempty values are pairwise disjoint and cover Ω. Its occupied sites and rank are

$$\operatorname{supp} P = \{q : P(q) \neq \varnothing\}, \qquad \operatorname{rank} P = |\operatorname{supp} P|. \tag{2.1}$$

The nonempty values are packets, distinguished by their literal atom sets. The site maps are

$$p(q) = q + 1, \qquad d(1) = 0, \quad d(q) = q \quad (q \neq 1). \tag{2.2}$$

They act by pushforward union:

$$(a_{\#}P)(y) = \bigcup_{q:a(q)=y} P(q), \qquad a \in \{p, d\}. \tag{2.3}$$

Words are chronological: uv means execute u and then v. Write $P \cdot w$ for the successor. A **fusion** occurs exactly when a d step has both sites $0, 1$ occupied. Its complete record is

$$\rho = (d, 0, 1, U, V, 0, U \cup V), \qquad U = P(0), \quad V = P(1). \tag{2.4}$$

The order of U, V is part of the record. Let $\mathcal{H}_P(w)$ be the list of all such records in chronological order. Nonfusing letters add nothing to the list. Every fusion reduces rank by one, while every other step preserves rank. Thus an execution from rank t to rank s has exactly $t - s$ recorded events.

For an integer a, let $[a]_n$ be its least nonnegative residue. All powers of p below have nonnegative exponents; each letter in them is charged. In particular, an inverse rotation is represented by a positive power, not supplied as another generator.

Fix an initial placement P_0 of rank t. List its packets in positive physical cyclic order as $B_0, \ldots, B_{t-1}$, with indices understood cyclically. A cyclic block of initial packets is a union of consecutive B_i in that order. No numerical cyclic order on Ω is required.

Lemma 2.1. Cyclic Packet Order

In every execution from P_0, each packet is a cyclic block of initial packets, and the physical cyclic order of the current packets agrees with their induced cyclic order. Every effective merger combines ordered cyclic neighbors. At rank two both ordered pairs are possible cyclic neighbor pairs.

Proof. The assertion holds initially. A p step rotates the physical positions and changes no packet. A nonfusing d either fixes the placement or moves the packet at site 1 into the adjacent empty site 0. It crosses no occupied site and preserves cyclic packet order. A fusing d replaces the packets at adjacent sites $0, 1$ by their union. They are cyclic neighbors in that ordered direction, so their union is again a cyclic block and the remaining packet order is preserved. Induction on the word proves the assertion. When only two packets remain, each is the next packet after the other in the cyclic list. □

An **admissible schedule** from P_0 is a finite sequence of records of the form (2.4), constructed by repeatedly choosing two distinct ordered cyclic neighbors in the current packet list and replacing them by their union. At rank one no further merger is allowed. The initial list is the list of packets of P_0. For a schedule ending at rank s, a **compatible endpoint** is any placement of its final s packets at distinct sites in the induced cyclic order. These conditions describe a formal schedule and endpoint without assuming an execution word for them.

3 Fusion-Free Routing

Lemma 3.1. A Batch Through Consecutive Holes

Suppose a packet V is at site b, and the sites $b-1,\dots,b-f$ are holes, where $1 \le f \le n-2$. The word

$$\mathsf{B}_{b,f} = p^{[1-b]_n} d (pd)^{f-1} p^{[b-f]_n} \tag{3.1}$$

moves V to $b-f$, restores every other packet to its original site, and produces no fusion. Its length is at most $2n+f$.

Proof. The prefix sends V to site 1 and the preceding original hole $b-1$ to site 0. The first d moves V to 0. Each subsequent p returns V to site 1, and the next d moves it into the next preceding hole. Before the kth d, the site at port 0 corresponds to the original site $b-k$, for $1 \le k \le f$. Thus no one of these steps fuses packets or moves another packet by d.

All other packets receive the total rotation

$$[1-b]_n + (f-1) + [b-f]_n \equiv 0 \pmod{n}.$$

They therefore return to their starting sites. The last d leaves V at 0, and the suffix puts it at $b-f$. If $f \ge 2$, the two outer exponents sum to a nonnegative integer congruent to $1-f$ modulo n and at most $2n-2$; hence $[1-b]_n + [b-f]_n \le 2n+1-f$. Adding the $2f-1$ intervening letters gives length at most $2n+f$. If $f=1$, the outer exponents sum to 0 or n, giving length 1 or $n+1$. □

A nonempty use of (3.1) is called one batch and accounts for f unit hole transfers. This terminology counts actual packet slides, not letters of a new alphabet.

Lemma 3.2. One Ordered Merger

Let U, V be ordered cyclic neighbors at sites a, b of a placement with at least two packets. Set $f = [b-a-1]_n$. There is a word that produces exactly the event $(d, 0, 1, U, V, 0, U \cup V)$, with no earlier fusion, using at most one batch, f unit hole transfers and $3n+f$ letters.

Proof. All f sites between a and b are holes. If $f > 0$, apply Lemma 3.1 to V; the rank assumption ensures $f \le n-2$. It moves V to $a+1$ and restores every other packet, including U. For $f=0$ no batch is needed. Rotate by $p^{[-a]_n}$ and execute one d. The parents now occupy ports $0, 1$ in the requested order, so the event is exactly the specified tuple. The batch costs at most $2n+f$, and the final rotation and fusion cost at most n. When only two packets remain, apply this construction separately to (U, V) and (V, U). Both are ordered cyclic neighbors, and the two resulting records remain distinct even though their result sets coincide. □

Proposition 3.3. Same-Order Placement

Suppose $P, P^\star$ have the same s named packets in the same cyclic order. A fusion-free word takes P to $P^\star$, using at most

$$K_{\text{place}} \le s-1, \qquad M_{\text{place}} \le (n-s)(s-1), \tag{3.2}$$

and

$$H(n,s) = (3n-s)(s-1) + (n-1) \tag{3.3}$$

letters. For $s = 1$ it uses no batches and at most $n-1$ rotations.

Proof. Handle $s = 1$ by rotation. For $s \ge 2$, fix a cyclic naming $C_0, \dots, C_{s-1}$ of the packets. Let g_i and $g_i^\star$ be the numbers of holes after C_i and before C_{i+1} in the two placements. Both vectors sum to $h = n - s$.

Choose nonnegative integral flows satisfying

$$g_i^\star = g_i + f_{i-1} - f_i. \tag{3.4}$$

For example, form successive prefix sums of $g_i - g_i^\star$ around the cycle and subtract their minimum. Their total increment is zero, so this gives (3.4) and at least one zero-flow edge b. Moreover $f_i \le h$: relative to an edge where the minimum occurs, f_i is the difference of the current and target gap masses on one cyclic interval, each between 0 and h.

Starting after b, process edges $b+1, \dots, b+s-1$ in cyclic order. At edge i move C_{i+1} backward through f_i holes by Lemma 3.1, omitting the step when $f_i = 0$. The preceding transfer has added f_{i-1} holes to gap i; later transfers have not yet touched it. Its current size is therefore

$$g_i + f_{i-1} = g_i^\star + f_i \ge f_i. \tag{3.5}$$

Every batch is enabled on the actual placement left by its predecessor. It decreases gap i by f_i and increases gap $i+1$ by that amount, without moving another packet or creating a fusion. When the last edge has been processed, (3.4) also fixes the zero-flow gap b, so all gaps equal their target values.

There are at most $s-1$ nonzero flows and their sum is at most $h(s-1)$. The batches cost at most $2n(s-1) + h(s-1)$. A final rotation sends C_0 to its prescribed site, costing at most $n-1$ letters. Equal gap vectors then fix every other named packet at its prescribed site as well. This proves literal placement equality and (3.2)–(3.3). If $s = n$, both gap vectors and every normalized flow are zero, so the construction uses only the final rotation. □

4 Joint Realization and Quadratic Normal Form

Theorem 4.1. Log and Endpoint Realization

Fix any initial placement P_0 of rank t, and let $1 \le s \le t \le n$. A structured list $\mathcal{H}$ and a placement $P^\star$ of rank s are jointly attainable from P_0 if and only if $\mathcal{H}$ is an admissible schedule and $P^\star$ is compatible with it. When these conditions hold, one word W satisfies

$$\mathcal{H}_{P_0}(W) = \mathcal{H}, \qquad P_0 \cdot W = P^\star, \tag{4.1}$$

with

$$\begin{aligned} |W| \le B(n,t,s) := 3n(t-s) + (n-t)(t-s) + \binom{t-s}{2} \\ + (3n-s)(s-1) + (n-1). \end{aligned} \tag{4.2}$$

Proof. Necessity follows from Lemma 2.1 and the exact event definition. For sufficiency, let $d = t - s$ be the number of schedule records and write $\mathcal{H} = (\rho_1, \ldots, \rho_d)$. Starting at P_0, replay the records by Lemma 3.2. After j events maintain the invariant that the actual log is $(\rho_1, \ldots, \rho_j)$ and the actual source-block partition is the abstract schedule partition after those events. Physical positions are not part of this induction invariant.

The next recorded parents are present by that partition equality. Their ordered adjacency in the abstract cyclic list is also their ordered adjacency in the actual placement, by Lemma 2.1. Lemma 3.2 therefore applies to the current concrete placement, appending exactly the next record and making the required union. The induction continues from the placement reached by the preceding record.

After the d events, the current and target placements have the same packets in the same cyclic order. Proposition 3.3 restores $P^\star$ without adding an event. The concatenated word proves both equalities in (4.1). If $t = s$, the replay stage is empty and only Proposition 3.3 is used. If $s = 1$, its final stage is just the rotation of the single packet, including the case $t = s = 1$.

Initially there are $h = n - t$ holes. Before event $j + 1$ there are $h + j$ holes, so its parent gap has length $f_j \le h + j$. Its route costs at most $3n + f_j$. Thus the complete fusion stage costs at most

$$\sum_{j=0}^{d-1} (3n + f_j) \le 3nd + hd + \binom{d}{2}. \tag{4.3}$$

Adding $H(n, s)$ proves (4.2), including an empty schedule. □

Corollary 4.2. Uniform Bound and Routing Budgets

The word in Theorem 4.1 can be chosen with

$$|W| \le \frac{7}{2} n(n-1), \qquad K \le t - 1, \tag{4.4}$$

and

$$M \le (t - s)(n - t) + \binom{t - s}{2} + (n - s)(s - 1). \tag{4.5}$$

For a fully occupied start, $K \le n - 2$ when $s < n$; when $s = n$, a rotation of at most $n - 1$ letters suffices.

Proof. The replay uses at most $t - s$ batches and the final placement at most $s - 1$, proving the bound on K. Summing the parent gaps in (4.3) and the terminal transfer bound proves (4.5).

Define

$$Q(n, s) = 3n(n - s) + \binom{n - s}{2} + H(n, s).$$

With $h = n - t$, expansion gives

$$Q(n, s) - B(n, t, s) = 3nh + \binom{h}{2} \ge 0, \qquad Q(n, s + 1) - Q(n, s) = -(n + s - 1) < 0. \tag{4.6}$$

It follows that $B(n, t, s) \le Q(n, 1) = \frac{7}{2} n(n - 1)$. If $t = n$ and $s < n$, the first merger has zero parent gap, removing one batch from the bound. If $t = s = n$, the compatible endpoint is a rotation of the start. Sparse zero-merger targets are covered by Proposition 3.3. □

Corollary 4.3. Complete Effective-Log Compression

For every word w from P_0, there is a word W of length at most $B(n,t,s)$, where $s = \operatorname{rank}(P_0 \cdot w)$, with

$$\big(\mathcal{H}_{P_0}(W), P_0 \cdot W\big) = \big(\mathcal{H}_{P_0}(w), P_0 \cdot w\big). \tag{4.7}$$

Proof. Its actual log is admissible and its endpoint compatible, so apply Theorem 4.1. □

Every observation determined by the pair in (4.7) is preserved. The recorded target consists of effective events and the literal endpoint; letter-indexed null outputs and intermediate placements are outside it.

5 Log Counts and Forgetful Fibres

For $t = 1$ the only schedule is empty. For $t \geq 2$, mark the t original directed boundaries between successive B_i. Until rank one is reached, every current packet corresponds to one interval of this initial packet circle. Merging ordered neighbors deletes the original boundary from the left parent to the right parent.

Lemma 5.1. Boundary-Deletion Encoding

Schedules ending at rank s are in bijection with ordered lists of $t - s$ distinct initial boundaries.

Proof. With at least two current intervals, each remaining directed boundary has a distinct interval on either side. Deleting it merges those two intervals in the boundary's order, so every ordered list of $t - s$ distinct boundaries gives a schedule.

For the inverse, retain the initial boundary labels on the current interval circle. The complete ordered parents in the next record identify the unique boundary directed from the last initial packet of the left interval to the first initial packet of the right interval. Delete this label and continue. At rank two the two possible parent orders select the two different remaining boundaries, although both result sets are the full atom set. Induction on the record list proves that encoding and decoding are inverse. At rank one the last undeleted boundary is not visible in the endpoint, but is recovered as the unique label absent from the deletion list. The $t = s = 1$ case is the unique empty list. □

Theorem 5.2. Rank-Refined Joint Count

From any fixed rank-t start, for $1 \leq s \leq t$,

$$\#\{\mathcal{H} : \text{terminal rank } s\} = \frac{t!}{s!}, \qquad \#\{P^\star \mid \mathcal{H}\} = s\binom{n}{s}, \tag{5.1}$$

and therefore

$$\#\{(\mathcal{H}, P^\star) : \operatorname{rank} P^\star = s\} = \frac{t!}{s!}\, s\binom{n}{s}. \tag{5.2}$$

Proof. Lemma 5.1 gives $t!/s!$ boundary lists. Fix one log and its s resulting named packets. Choose an s-element occupied-site set in $\binom{n}{s}$ ways and choose which packet occupies its least site in s ways. The cyclic order fixes every other packet at every other occupied site. Conversely the literal placement recovers both choices. Theorem 4.1 realizes each choice for this very log. This proves both fibre cardinality and the product count. For $s = 1$ the occupied-site choice is one of the n phases. □

Proposition 5.3. Fibres over Literal Endpoints

At rank $s \geq 2$ there are

$$s\binom{n}{s}\binom{t}{s} \tag{5.3}$$

attainable literal endpoints. Every such endpoint admits exactly $(t-s)!$ complete logs. At rank one there are n endpoints, each admitting $t!$ complete logs.

Proof. If $s \geq 2$, the endpoint's source partition determines its s surviving initial boundaries. The remaining $t-s$ boundaries can be deleted in any order; Lemma 5.1 and Theorem 4.1 make these exactly the logs over that endpoint. Choosing the surviving boundaries gives $\binom{t}{s}$ source partitions, and each has $s\binom{n}{s}$ placements. At rank one no surviving boundary is identified by the full atom set. There are t choices of that boundary and $(t-1)!$ deletion orders, giving $t!$ logs over each endpoint. This also covers $t = 1$. □

For a singleton start $t = n$, summing (5.2) over rank gives

$$n! \sum_{s=1}^{n} \frac{\binom{n}{s}}{(s-1)!}. \tag{5.4}$$

The rank-one contribution alone is $n\,n!$. Equation (5.4) counts distinct complete logs paired with literal endpoints.

6 A Specified Quadratic Optimum

Now take $\Omega = Q$ and the singleton start $P_{\text{seed}}(q) = \{q\}$. Prescribe the all-right log

$$\mathcal{H}_R = (\rho_1, \ldots, \rho_{n-1}), \qquad \rho_i = (d, 0, 1, \{0, \ldots, i-1\}, \{i\}, 0, \{0, \ldots, i\}), \tag{6.1}$$

and prescribe the final packet Q at site $n-1$. Let $L_R(n)$ be the minimum length realizing this log and endpoint.

Lemma 6.1. An Explicit Realization

The chronological word

$$W_n = d\left[\prod_{j=1}^{n-2} p^{n-j} d(pd)^j\right] p^{n-1} \tag{6.2}$$

realizes the target and has length $\frac{3}{2}n(n-1) - 1$.

Proof. The initial d creates $\{0, 1\}$ at 0. Before stage j, the carrier is $\{0, \ldots, j\}$ at 0, the sites $1, \ldots, j$ are holes, and the remaining singletons occupy their original sites. After p^{n-j}, singleton $j+1$ is at 1 and site 0 is empty. The first d slides that singleton to 0. The next j copies of pd slide it through the holes; on the last copy the carrier has returned to 0, so the d fuses the requested ordered parents. There are j nonfusing slides and one fusion in this stage. Every other packet receives a full n rotations and no slide, so its original site is restored. This proves the stage invariant.

After the last stage the full packet is at 0, and the suffix puts it at $n-1$. The stages cost $n+j+1$ letters each, so

$$|W_n| = 1 + \sum_{j=1}^{n-2}(n+j+1) + (n-1) = \frac{3}{2}n(n-1) - 1. \tag{6.3}$$

The number of rotations is $P_{\min} = n(n-1) - 1$, and the number of nonfusing slides is $T = (n-1)(n-2)/2$. □

We prove optimality by accounting separately for rotations and slides. Because (6.1) prescribes the complete fusion log, every realizing word has exactly those $n-1$ fusions: no additional outside-only fusion is allowed. Consequently every packet not yet absorbed into the carrier remains its original singleton. Every other d step is then an external-singleton slide, a carrier backslide from 1 to 0, or a step that leaves the placement unchanged. Write their counts as S, B, Z, respectively. Let B_m count carrier backslides while exactly m external singletons remain; thus $B = \sum_{m=0}^{n-2} B_m$.

Lemma 6.2. Rotation Cost of a Backslide

For any realizing word w,

$$N_p(w) \geq P_{\min} + B. \tag{6.4}$$

Proof. Before the first fusion, the placement is fully occupied, so the prefix consists of rotations. The first ordered parents in (6.1) force their number to be a multiple of n.

Between consecutive specified fusions, the carrier starts and ends at 0. Each rotation increments its site and each backslide decrements it, so $N_p - B$ on that interval is a multiple of n. Immediately after a fusion, and immediately after a carrier backslide, the carrier is at 0 and site 1 is empty. At least one rotation is needed before the next backslide. After the last backslide, or after the preceding fusion if there was no backslide, at least one further rotation is needed before the next specified fusion. A d cannot populate the empty site 1. Consequently $N_p \geq B + 1$ on the interval, making the multiple of n strictly positive. It follows that $N_p \geq n + B$ on each of the $n - 2$ inter-fusion intervals.

After the last fusion only the carrier remains. The same site accounting gives $N_p - B \equiv n - 1 \pmod{n}$ on the tail. Every backslide requires a preceding rotation, so this difference is nonnegative and hence at least $n - 1$. Adding the intervals proves (6.4). □

Lemma 6.3. Hole-Potential Identity

The external slides satisfy

$$S = T + \sum_{m=1}^{n-2} mB_m, \qquad T = \frac{(n-1)(n-2)}{2}. \tag{6.5}$$

Proof. While m external singletons remain, anchor their cyclic order at the carrier and let $g_0, \ldots, g_m$ be the successive hole gaps. In particular g_0 follows the carrier. Set

$$\Phi_m(g) = \sum_{i=0}^{m} (m - i) g_i. \tag{6.6}$$

Rotations and unchanged d steps preserve this potential. An external singleton slide transfers a hole from its preceding gap to its following gap, decreasing Φ_m by one. A carrier backslide transfers a hole from g_m, whose coefficient is zero, to g_0, whose coefficient is m, increasing the potential by m.

At a prescribed right fusion the gap g_0 is zero. After the fusion the new gaps are $(g_1 + 1, g_2, \ldots, g_m)$: the vacated site 1 adds one hole immediately after the enlarged carrier. Therefore

$$\Phi_{m-1}(g_1 + 1, g_2, \ldots, g_m) - \Phi_m(0, g_1, \ldots, g_m) = m - 1. \tag{6.7}$$

Immediately after registration $m = n - 2$, with gap vector $(1, 0, \ldots, 0)$ and potential $n - 2$. At rank one the potential is zero. Summing all changes yields

$$0 = (n - 2) - S + \sum_{m=1}^{n-2} mB_m + \sum_{m=1}^{n-2} (m - 1).$$

The two terms independent of backslides sum to T, proving (6.5). The rank-one backslides B_0 have no effect on this identity. □

Theorem 6.4. Exact Optimum and Excess Cost

The specified target has exact minimum

$$L_R(n) = \frac{3}{2}n(n-1) - 1. \tag{6.8}$$

More precisely, put $R = N_p(w) - P_{\min} - B \geq 0$. Every realizing word satisfies

$$|w| - L_R(n) = R + 2B_0 + \sum_{m=1}^{n-2}(m+2)B_m + Z. \tag{6.9}$$

Proof. There are $n-1$ fusing letters, so $|w| = N_p + (n-1) + S + B + Z$. Substitute (6.4)–(6.5) and $N_p = P_{\min} + B + R$. The constant part is $P_{\min} + (n-1) + T = \frac{3}{2}n(n-1) - 1$, and the remaining terms give (6.9). They are all nonnegative. Lemma 6.1 attains the bound. □

The identity specifies the extra cost of certain detours: a carrier backslide with m external packets forces at least $m+2$ additional letters, and an unchanged d forces one. These costs are relative to the specified log and endpoint.

Corollary 6.5. Uniqueness of the Shortest Word

The only shortest word for this target is W_n in (6.2).

Proof. Equality in (6.9) excludes carrier backslides, unchanged d steps and rotation excess. The resulting stage-by-stage forcing argument is given in Appendix A. □

Let $D_n^{\log}$ be the maximum, over all jointly attainable complete logs and literal endpoints from the singleton start, of their minimum realization length. Theorem 5.2 makes this a maximum over a finite set. Theorems 4.1 and 6.4 give

$$\frac{3}{2}n(n-1) - 1 \leq D_n^{\log} \leq \frac{7}{2}n(n-1). \tag{6.10}$$

Thus $D_n^{\log}$ has quadratic order. The exact maximum is not needed for this conclusion.

7 Forgetting History Can Change the Cost Order

Consider the much shorter word

$$V_n = d(pd)^{n-2}p^{n-1}. \tag{7.1}$$

After the first d, every pd brings the preceding singleton to port 0 and the carrier to port 1, then fuses them. It reaches the full packet at 0 and finally at $n-1$. Hence it has exactly the same literal endpoint as W_n, with length $3n-4$. Its ordered merger log differs from (6.1).

Write E_n for the minimum length reaching this endpoint when history is unrestricted. A rank-n start requires $n-1$ effective mergers to reach rank one, so $n-1 \leq E_n \leq 3n-4$ and $E_n = \Theta(n)$. Meanwhile Theorem 6.4 implies

$$\frac{L_R(n)}{E_n} \geq \frac{\frac{3}{2}n(n-1) - 1}{3n-4} \sim \frac{n}{2}. \tag{7.2}$$

This is an unbounded distortion under the map that forgets the complete log. It compares two optimization problems in the same literal alphabet from the same source, with the same terminal placement. The quadratic cost belongs to the prescribed chronological mergers, rather than to that endpoint alone.

8 Registered Observations and Finite Checks

A registered observation can be read from a complete log by fixing an atom set F, retaining the first event whose result is exactly F as its origin, and then retaining later events in which the current packet containing F is a parent. The final placed partition supplies the current carrier and all outside packets. This recovers the origin and ordered selected absorption chain used in registered-summary classifications (Chen, 2026a,b).

Corollary 8.1. Compression of Attainable Registered Summaries

Every registered summary attained from a singleton start has a word realization of length at most $\frac{7}{2}n(n-1)$ preserving its full origin, selected chain and literal terminal partition.

Proof. Apply Corollary 4.3 to one word attaining the summary. Its complete log and endpoint determine all the specified reads. □

Finite-state enumeration checks the construction from singleton starts for $3 \le n \le 6$.

Table 1. Joint complete-log/endpoint objects from singleton starts, including the initial empty log.

Sites n	Joint log/endpoint objects
3	39
4	292
5	2,505
6	24,306

An arbitrary-start enumeration checks 65,992 joint log/endpoint objects across 36 layouts of disjoint two-atom packets and all initial ranks for $3 \le n \le 6$. It compares event lists, literal endpoints, and the $K, M, B(n, t, s)$ budgets. These finite instances corroborate the formulas; the all-parameter conclusions follow from the proofs above.

9 Conclusion

The schedule/placement characterization separates joint feasibility from the cost of retaining a particular history. The all-right example makes that distinction sharp in order: its prescribed log requires quadratic length, while its endpoint admits a linear word. The routing bounds give short representatives for every attainable log/endpoint pair from an arbitrary named-packet start.

Appendix A Forcing the Unique Shortest Word

Let w attain equality in (6.8). By (6.9), $R = B = Z = 0$. In the proof of Lemma 6.2 every interval contributes a nonnegative rotation excess. Thus there are no rotations before registration, exactly n rotations between consecutive fusions, and exactly $n-1$ rotations after the last fusion. Registration is the first letter d.

For singleton i, let M_i be its number of external slides before it is absorbed. At its fusion it occupies port 1. The cumulative number of rotations is a multiple of n, and its initial site is i, so

$$M_i \equiv i - 1 \pmod{n}, \qquad M_i \ge i - 1. \tag{A.1}$$

All nonfusing letters are such slides, and (6.5) gives $\sum_{i=2}^{n-1} M_i = T = \sum_{i=2}^{n-1}(i-1)$. Therefore $M_i = i - 1$ for every i.

We now force stage j by induction, starting from the canonical state in Lemma 6.1. Before its first d, let t be the number of rotations. At $t = 0$, port 1 is empty, so a d would be unchanged. At $t = 1$, the carrier is at port 1 and an outside singleton is at port 0, so a d would make a

forbidden left merger. For $2 \le t < n - j$, both ports are occupied by outside singletons, giving an unrecorded outside merger. Thus no d can occur before $t = n - j$.

At that time singleton $j + 1$ first reaches port 1 over a hole at 0. It must slide then. Otherwise it cannot reach port 1 again within the stage's total n rotations, and cannot be its prescribed final right parent. Following this first slide, the remaining j rotations must each be followed by d: if a slide is skipped, that singleton again cannot return to port 1 within the remaining rotation budget. The first $j - 1$ such letters are slides and the last is the required fusion. The forced stage is therefore $p^{n-j}d(pd)^j$.

The other packets receive n rotations and no slide during this forced stage, restoring the canonical state for stage $j + 1$. After the last fusion no backslide or unchanged d is permitted, so the tail is exactly p^{n-1}. This proves $w = W_n$.

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